

Mitigating Ocean Acidification and Plastic Contamination through Biogenic Infrastructure

Word Count: 687

In 2026, a third-generation oyster farmer in Lau Fau Shan pulled up his nets and found nothing but sludge and debris where massive molluscs once grew. The water was too acidic, the heat too unrelenting, and his family's legacy was disappearing in a single season. The plastic legacy of our past had caught up to our dinner plates. He didn't just lose his income, but his connection to the sea and a link to his environment.

The Pearl River Delta was at a critical point of ecological threshold. For decades, the shallow, brackish waters of Deep Bay served as a depositional zone for synthetic pollutants and industrial excess. Locally, plastics now make up 18% of our daily waste, and water samples found that 60% contained plastic microbeads, identifying concentrations of up to 380,000 beads per square kilometre (So et al., 2018). These particles, largely originating from personal care products, often bypass treatment systems to enter the marine environment. By 2025, research confirmed that ocean acidification had surpassed safe planetary boundaries, threatening global marine stability (Findlay et al., 2025). Deep bay required an immediate, holistic intervention to mitigate chemical instability and rising sea levels.

The current method adopted uses rigid concrete-based infrastructure, such as sea walls that are structurally static and unable to adapt to environmental shifts. Instead, we can implement nature-based engineering, such as biogenic eco-shorelines utilising reefs of Native Hong Kong Oysters (*Magallana hongkongensis*), which act as chemical buffers. As older shells within the reef naturally erode, they release carbonate in the water, neutralising the carbonic

acid formed by atmospheric carbon dioxide (NOAA Fisheries, 2023). The process allows for biodiversity to be maintained despite the acidification of the open South China Sea.

The solution is also efficient, as a single oyster can filter up to 200 litres of water a day; the oysters can remove and store pollutants. Through the rigorous filtration, they remove microbeads and other pollutants from the waters and incorporate them into pseudofeces (Thushari & Senevirathna, 2020). These particles are then buried within the anaerobic layers of the reef bed, preventing them from entering the food chain and thereby purifying the waters. This biological filtration is enhanced by mangroves, which reduce water velocity, ensuring that plastic debris is captured before reaching the open sea (WWF Hong Kong, 2025). Unlike concrete, these reefs grow vertically in response to sea-level rise (CEDD, 2026).

Additionally, the transition requires a restructuring of the current view between green infrastructure and urban development. This is because traditional infrastructure built with concrete is often prioritised, as the value is easily calculated in short-term construction projects. We often externalise the environmental costs of production and prioritise short-term resource depletion over long-term stability. Instead, development should be regenerative, where biological systems are valued as functional infrastructure. This encourages infrastructure that does not require the constant energy burden and maintenance of mechanical systems. Through legally classifying these reefs as essential defence systems, we can protect the local economy from bio-accumulated toxins while providing storm surge protection for coastal residents.

The use of these oyster reefs also allows for thermal heat management of the city, as Hong Kong, a coastal city, suffers from the effects of urban heat island, where concrete and asphalt surfaces absorb and re-radiate heat, raising local temperatures compared to rural surroundings. Conventional concrete sea walls contribute to this effect due to their high thermal mass. In contrast, biogenic eco-shorlines such as oyster reefs and mangrove vegetation facilitate evaporative cooling. Mangroves release water vapour through transpiration, and the irregular surface area of the reef increases evaporation rates. Thereby lowering the ambient temperature of the coastal regions.

Hong Kong's primary defence should no longer rely on the rigidity of concrete. The example of oyster reef restoration demonstrates that providing a structural foundation allows the ecosystem to provide necessary coastal protection. By moving away from solutions that may cause environmental depletion and shifting towards deep ecological integration, we ensure that the legacy of Lau Fau Shan survives as a functional shield for the future. We have the opportunity to move beyond merely surviving environmental shifts to actively maintaining the ecological systems we inhabit.

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